

Coding & Solution

Date	04 july 2026
TeamID	XXXXXX
ProjectName	Human Development Index(HDI) Predictor
MaximumMarks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
RepositoryLink/ URL	https://github.com/itsnaseersyed/Human-Development-Index
ProgrammingLanguage(s)	Python
Framework(s) Used	Streamlit, Pandas, NumPy, Scikit-learn, Matplotlib
KeyFeatures Implemented	Human Development Index (HDI) data preprocessing, interactive dashboard, country-wise HDI analysis, trend visualization, comparative analysis, filtering and search functionality, statistical insights, and data-driven reports.
Pending/Incomplete Features	Human Development Index (HDI) data preprocessing, interactive dashboard, country-wise HDI analysis, trend visualization, comparative analysis, filtering and search functionality, statistical insights, and data-driven reports.
Setup/Run Instructions	Install the required dependencies using <code>pip install -r requirements.txt</code>. Launch the application using <code>streamlit run app.py</code>, then access the dashboard through the local browser.

Code Quality Checklist

S.No	Criteria	Status(Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes/Comments:

The Human Development Index (HDI) project is designed following clean coding practices and a modular architecture to improve readability, maintainability, and scalability. The application processes HDI data efficiently, provides interactive visualizations and comparative analysis, and supports future enhancements such as predictive analytics and integration with real-time data sources. The source code is version-controlled to facilitate collaboration and project maintenance.